Simulation Analysis

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1 Introduction

Blocked cluster randomized controlled trials (CRTs) are a widely used experimental evaluation design in which entire groups (clusters) such as schools or classrooms are randomly assigned to treatment or control conditions within defined strata or blocks. The effectiveness of the program or treatment is then typically quantified using the average treatment effect (ATE). However, within this relatively straightforward quantity,

researchers must make a variety of choices. Researchers need to decide whether to target a person-weighted average or cluster-weighted average; they need to decide whether they are targeting an estimate just for the clusters in the study (finite sample) or an estimate representative of some broader population (superpopulation); and they need to decide from among different methods or estimators that can be used to estimate the ATE.

This paper contains simulations that complement research examining empirically the implications of different choices researchers make when performing CRTs (Miratrix et al., 2025). Simulations, unlike empirical analyses, allow us to compare estimated quantities to their true values, which we cannot observe in empirical contexts. This in turn allows us to evaluate how well different statistical methods perform. Additionally, we can systematically vary data conditions and identify which factors impact method performance. # Research Questions

With this in mind, these simulations answer the following research questions:

1. Are any of the estimators ever substantially biased? When?
2. Are any of the estimators generally more precise than others? When?
3. Which estimators produce the best estimates of the treatment effect standard error?
4. Do any of the estimated standard errors target finite versus super-population uncertainty?
5. Why does the range in the estimates produced by the different estimators get larger as the degrees of freedom get smaller? Is the cause estimator bias or estimator instability?

2 The Simulation

We ran a series of multifactor simulations, which are simulations where we systematically vary a set of factors to understand how the factors influence the performance of the various estimators, and how the factors interact with each other. In general we are close to full multi-factorial designs, where we vary all factors independently, but we do drop combinations that do not make sense (in particular, we drop scenarios that call for correlations between quantities that do not vary).

2.1 The data generation parameters

The main simulation is a fully factored experiment with the following data generating parameters:

Table 1: Simulation Data Generating Parameters

Parameter	Value(s)
Parameters fixed across all scenarios	
Number of Blocks	5
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Block-Level (Varies across scenarios)	
Block Type (blk_type)	Same (Sam), Reasonable (Res), Messy (Mss)
Average Number of Clusters per Block (J.bar)	5 (S), 10 (M), 20 (L)
Proportion of Clusters Treated (p.bar)	.5 (Bal), .7 (UnB)
Cluster-Level (Varies across scenarios)	
Correlation of Impact and Size of Cluster (cor_J_tau)	0 (0), .8 (+)
Cluster Level Treatment Effect Standard Deviation (sd.clu)	0 (0), .3 (+)
Cluster Level ICC (ICC.clu)	0 (0), .2 (+)
Variation in cluster size, coefficient of variation (cv.clu)	0 (0), .33 (S), .6 (L)
Number of Cluster-Level Covariates (num.cov)	0 (0), 1 (S), 5 (L) [R^2 Always .5]

One of our factors is "block type," which is a set of seven parameters that define how similar or different the blocks are from each other. We did not independently vary these factors so we could control the total number of simulation scenarios. All code is public; others can easily adapt the code to run additional scenarios that are of interest. We have three levels of block type, which we call "Blocks All The Same," "Blocks All Reasonable," and "Blocks All Messy." The "Same" case is where all blocks are identical in their characteristics. The "Reasonable" case is what we might imagine we would see in actual practice. The "Messy" case pushes our simulation to the limits so we can see how various methods fail in extreme circumstance.

Table 2: Block-Type Parameters

Parameter	**Blocks All The Same**	**Blocks All Reasonable**	**Blocks All Messy**
Variation of number of clusters in the blocks, coefficient of variation	0 (none)	.33 (average)	.85 (big)
Variation across blocks in the average cluster size, coefficient of variation	0	.33	.33
Block-Level Average Treatment Effect Standard Deviation	0	.2	.6
Correlation Between Block Average Treatment Effect and Block Average Cluster Size	0	.3	.8
Correlation Between Block Average Cluster Size and Total Number of Clusters	0	.3	.8
Correlation Between Control-Side Cluster Average Outcome and Cluster Size	0	.3	.8
Correlation of Block's Proportion of Clusters Treated and Block Average Impact	0	.3	.8
Does the proportion of clusters treated vary across blocks?	No	Yes	Yes

Each scenario would have a specific combination of these parameters. We drop all scenarios with a correlation of a quantity that does not vary (e.g., if there is no treatment effect variation, then we drop scenarios with nonzero correlation between treatment effect and cluster size).

We also ran a separate set of simulations to evaluate how standard errors perform for fully interacted models in matched pairs studies. These simulations were also fully factored and the parameters for those simulations were as follows:

Table 3: Matched Simulation Data Generating Parameters

Parameter	Values
Block-Level Fixed	
Number of Blocks	20
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Proportion of Clusters Treated	.5
Average Number of Clusters per Block	2
How much do number clusters per block vary (coefficient of variation)?	0
Correlation Between Treatment Effect and Cluster Size	0
Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters	0
Correlation of Proportion of Clusters Treated per Block and Impact	0
Correlation Between Outcome Intercept and Cluster Size	0
Block-Level Varying	
Block-Level Average Treatment Effect Standard Deviation	0, .6, .2
Cluster-Level Fixed	
Number of Cluster-Level Covariates	0
Cluster-Level Varying	
Correlation of Impact and Size of Cluster	0, .8
Cluster Level Treatment Effect Standard Deviation	0, .3
Cluster Level ICC	0, .2
How much does the size of clusters vary (coefficient of variation)?	0, .33, .6

3 Data Generating Process

For most of our simulations, for each scenario defined by sample size, target correlation, predictiveness of covariates, and so forth, we used the following data generating process (DGP):

1. Randomly generate a single set of blocks based on the simulation scenario parameters. Each generated block has a set of characteristics such as the average treatment effect of the block, average cluster size, amount of cluster variation, and so forth.
2. Then, for each simulation iteration:
    a. Generate a sample of clusters within each block according to the block characteristics.
    b. Generate individual-level data within each cluster.
    c. Calculate the true ATEs (both cluster-weighted and individual-weighted) for the generated dataset. These are our target finite-sample estimands.
    d. Assign a proportion of clusters to treatment according to the target block-level proportion.

At this point we have a complete set of generated data. Next, for each simulation iteration we actually conduct a small number of iterations. For each iteration we:

a. Permute treatment across clusters within block to generate a new random assignment.
b. Calculate observed outcomes based on the generated data and the new treatment assignment.
c. Estimate the ATE and SE using all our methods.

The nested permutation approach means that for each fixed set of clusters and blocks, we can calculate finite sample performance. For example, the standard deviation of the ATE estimates within a given dataset across the permutations gives us the true finite sample standard error for that dataset. If we look at the standard deviation of the ATE estimates across all our datasets within a scenario, we have the true cluster-level superpopulation standard error. This standard error *does not* take into account uncertainty at the block level, as the block characteristics are all fixed. So, for example, if there is treatment variation across blocks, our true standard error will not reflect uncertainty due to which blocks were generated, only uncertainty due to which clusters and individuals were generated.

For our target estimands we can use, within each generated dataset, the finite sample estimand. We can also calculate the superpopulation estimand (both person-weighted and cluster-weighted) by averaging the finite sample estimands across all generated datasets within a scenario.

We next outline the data generation parameters that drive our DGP, and then give some further technical detail of how we implemented the above steps.

3.1 How the data are generated

For a given scenario, we start with a set of data generating parameters as described above.

Here is one of the more extreme scenarios from our main simulation:

Table 4: Simulation Scenario Parameters

variable	value
ATE	0.2
ICC.block	0.2
ICC.cluster	0.2
J.bar	10

variable	value
K	5
R2.cluster	0.5
cv.block	0.85
cv.cluster	0.33
cv.nbar	0.33
date.created	11/20/2025
empirical	TRUE
n.bar	20
nbar.J.cor	0.8
num.cov.cluster	5
num.cov.indiv	0
p.bar	0.7
proptx.impact.cor	0.8
sd.block	0.6
sd.cluster	0.3
size.impact.cor	0.8
size.impact.cor.cluster	0.8
size.intercept.cor	0.8
variable.p	TRUE

Then, to generate the individual blocks, we use a multivariate normal distributions to generate the block-level pairs of average control-side outcome and average treatment effect (in the case of treatment variation). Other block characteristics that are to vary across block (e.g., proportion of clusters treated) are then generated according to various specified distributions. Finally, we induce correlations between block level quantities as described below, if the relevant correlation parameters are nonzero.

When generating block mean outcomes and mean effects, we use the "empirical" setting to make sure our multivariate normal distributions for the amount of cross-block impact variation and so forth equals our set values. This mitigates worry that the simulation results might be dependent on *which* blocks we have generated, as we would expect all sets generated, given a specific set of parameters, would lead to the similar final results. That said, we generate these values to target *block averages*, but if we have variable numbers of clusters per block, the cluster-average effects can deviate from these targets quite substantially, especially if number of clusters and treatment effect are correlated.

Overall, however, the finite nature of the blocks should not be a large concern. In particular, because we have a large number of factors in our experiment, when we look at the main effect of a given factor, we will be averaging across many different scenarios with a different specific set of blocks for each scenario, so any dependencies on the specific set of blocks generated would be ameliorated.

Once generated, we store the block characteristics as a small dataset, which we call a "grid," with each row being a block, and each column a characteristic we need for generating the within-block data (number of clusters, average cluster size, etc.). To illustrate, here is the "grid" of block-level characteristics for the above scenario in our main simulation:

did	tau	alpha	J	p	n.bar	ICC.cluster	k.clu	R2.cluster	cv.cluster	cor.clu	k.indiv
1	-0.2	-0.6	2	0.5	12	0.2	5	0.5	0.3	0.8	0
2	0.6	0.0	8	0.8	15	0.2	5	0.5	0.3	0.8	0
3	1.1	0.3	11	0.8	23	0.2	5	0.5	0.3	0.8	0
4	-0.4	-0.3	3	0.7	15	0.2	5	0.5	0.3	0.8	0
5	0.0	0.5	7	0.6	14	0.2	5	0.5	0.3	0.8	0

and here is a set of descriptive features for data we would get from this fixed set of blocks:

```
## K=5  J.bar=17.20 (13.66, cv=0.79)    n.bar=29.03 (9.29 cv=0.32)  p.bar=0.71 (0.07)
## District stats:
##  block ATE=0.20 (0.60)   block ICC=0.45
##  cor(J,tau)=0.96     cor(ptx,tau)=0.40
##  cor(J,alpha)=0.87       cor(nbar, J)=0.95
## Cluster stats:
##  cluster ICC=0.20 (0.00)
##  avg cluster size = 29.1 (range 6--60, sd = 13.4, cv=0.46)
##  cor(n,tau) = 0.81 (across blocks)
## ATE information:
##  Within-block cluster ATE and variation: 0.20 (0.26)
##  Total cluster ATE: 0.56 (0.54)
##  Total individual ATE: 0.76 (0.50)
##
##   * Some stats estimated on 99957 generated observations across 3440 clusters.
```

The block ATE is the simple average of the block-level cluster-average ATEs. It is as we targeted. The bottom line are ATEs for a large dataset generated for this set of blocks: these are our approximated target superpopulation parameters for this scenario.

In this specific example, these are different from each other and from the block average. We also see fairly high correlations of all sorts of features of the data. The numbers in parentheses are the standard deviations of a given value.

Given our grid of block characteristics we then, for each block in turn, use a multilevel model data generating process (as described more fully in Miratrix et al.,2021) to generate the target number of clusters and individuals.

To keep everything in effect size units, we have a target within-block total variation of $1 - ICC_3$, where ICC_3 is the target intraclass correlation for the blocks. Note that we have $ICC_3 + ICC_2 + \frac{\sigma^2_\epsilon}{(1-R^2_1)} = 1$, with ICC_2 being the cluster-level ICC and σ^2_ϵ the individual-level residual variance.

3.2 How we correlate block-level quantities

Given two vectors of numbers, x and y, that we wish to correlate with target correlation ρ, we do the following:

1. Generate a bivariate standard normal with correlation ρ.
2. Re-order the y values so the pairs of the ranks of x and the ranks of the reordered y correspond to the pairs of ranks of the bivariate distribution.

If x and y are not normal, then this will not produce correlations that have the target correlation on average, but it will generally give high correlations when ρ is high, and negative correlations when ρ is negative. We are correlating based on the normal transform of the ranks, so this is somewhat akin to a rank correlation.

If we have multiple correlations, then we do these in sequence, holding one of the variables as fixed so we do not undo any of the other correlations. For example, say we need to generate a scenario with both a correlation between the number of clusters in a block and average impact in the block, and a correlation of proportion treated in a block and average impact in the block. To do this, we first generate the three vectors of number of clusters, average impact, and proportion treated. We then rearrange, as described above, the number of clusters so it correlates with average impact, and next rearrange the proportion of clusters treated so it correlates with average impact. This will induce a correlation between number of clusters in a block and proportion treated in a block, because they are both correlated with the block-level ATE.

3.3 How we generate distributions of sizes

We have three kinds of sizes to generate: number of clusters per block, number of individuals per cluster, and average size of cluster for each block.

For each of these we use a similar approach. We first specify the desired mean and coefficient of variation (cv). For example, we might want an average of 8 clusters per block with a cv of 0.75. This means the standard deviation of our values would be $sd = cv * mean = 0.75 * 8 = 6$.

We then have three possibilities: either the variance is less than the mean, in which case we can use a binomial distribution, or the variance is larger than the mean, in which case we can use a negative binomial distribution. If the variance is very close to the mean, we use a poisson distribution.

We use this approach for the average cluster size across blocks, even though the average cluster sizes could be fractional.

4 How to read the box plots in this paper

In this paper, we use a set of boxplots customized for our particular data. Below is a sample boxplot of the mean bias of an estimator relative to its target estimand across simulation runs for all methods and scenarios. The faint horizontal lines extend to the full range of the data. The bars mark the 2.5th and 97.5th percentiles, capturing 95% of the data. The boxes span the 10th and 90th percentiles, corresponding to the middle 80% of the data. Overall, the estimators exhibited negligible bias across a broad set of simulation conditions.

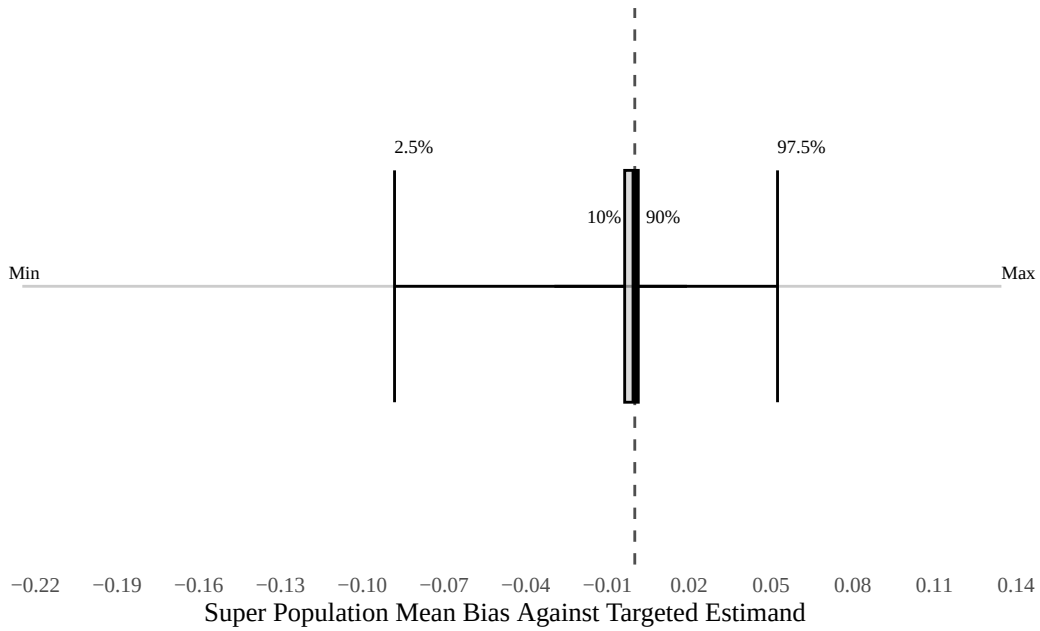


Figure 1: Sample boxplot for paper. This boxplot shows the superpopulation mean bias of each estimator relative to its target estimand across scenarios and methods. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

5 Are any of the estimators ever substantially biased? When?

In 80% of the scenarios tested, the magnitude of bias was less than 0.01 for all estimators (Figure 2), and the scenarios with bias require the confluence of multiple bias-causing factors (Figure 3). The estimators

that theory predicts are essentially unbiased do indeed have nearly no bias (or very low bias) across the full range of simulations. In some of the simulations, the person-weighted fully interacted estimator is predicated to have a bias with a magnitude of as large as .03, which the regression tree shows requires a correlation between the treatment effect and cluster size combined with a small number of clusters per block, large cluster size variation, and unbalanced treatment assingment (Figure 3). For the remaining estimators, except the multilevel fully interacted model, we observe bias as severe as -0.2 in some simulations, primarily driven by the combination of messy blocks and unbalanced treatment assignment of clusters. It's important to note here that the messy block condition was deliberately designed to push parameters to the extreme. Particularly the messy block condition has a .8 correlation between the proportion of each block assigned to treatment and the block-level treatment effect, which is particularly problematic for the multilevel estimators because it shifts the estimator weighting from cluster to person. Our simulations do not show substantial bias under the data generating conditions we expect to see in practice (i.e. the same and reasonable block types). The multilevel fully interacted model performed well with messy blocks and unbalanced treatment assignment, and instead was most biased when there was a combination of correlation between the number of clusters and the treatment effect, zero cluster-level ICC, and substantial variation in cluster size.

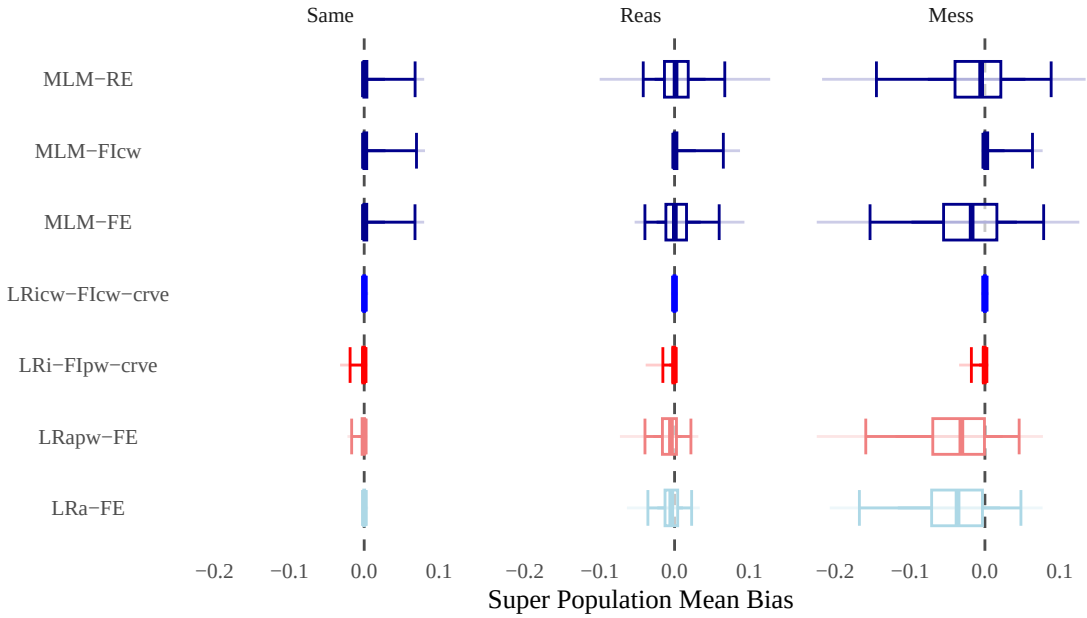


Figure 2: Boxplot of superpopulation mean bias of each estimator relative to its target estimand across scenarios, separated by method. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

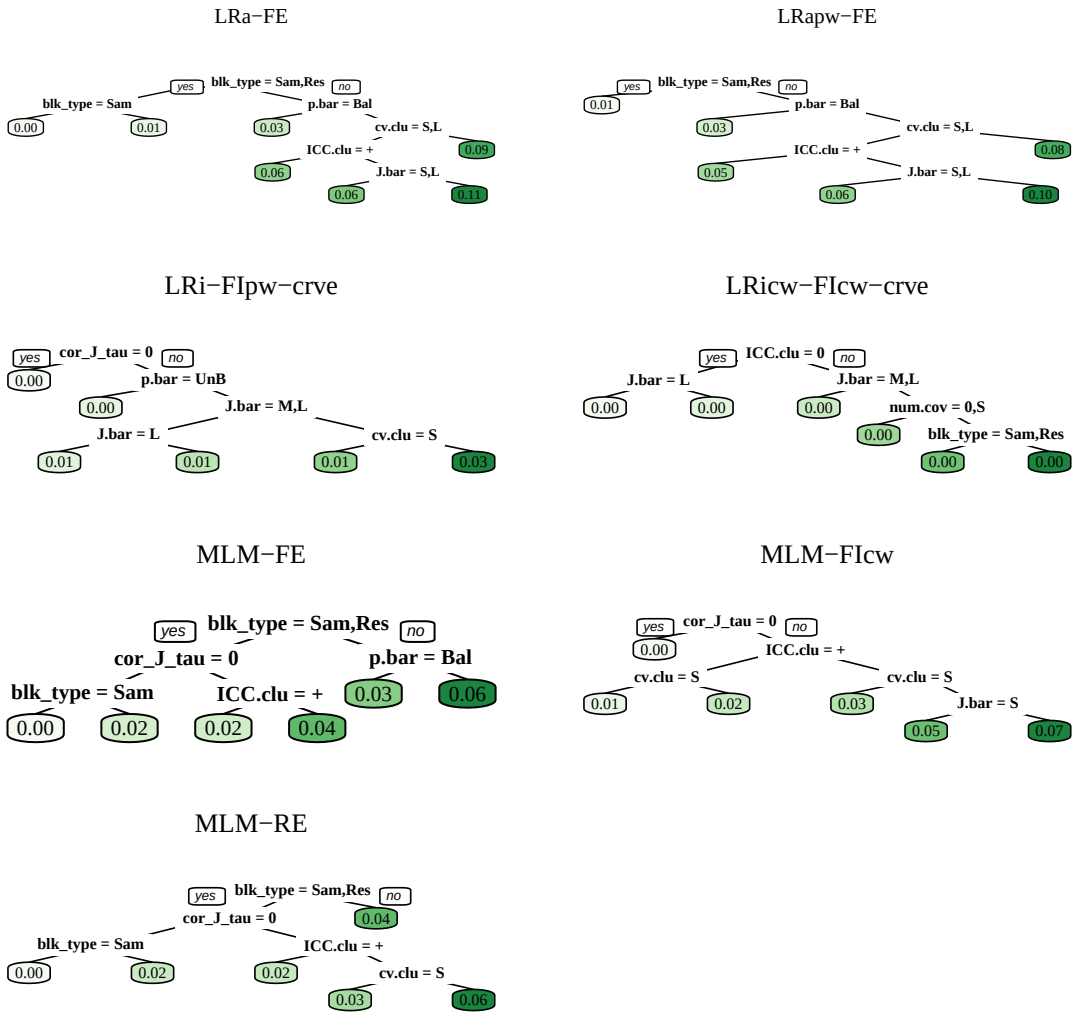


Figure 3: Regression tree by method predicting the magnitude of the mean bias under different data generating parameters

6 Are any of the estimators generally more precise than others? When?

In general, all the different estimators have comparable precision (Figure 4). Excluding the multilevel estimators, within a given scenario, all the estimators have standard errors that are at most 10% larger than the median across methods. The multilevel estimators become more precise than the other estimators when there is a combination of large variation in cluster size and zero cluster ICC; otherwise, their precision is close to that of the other estimators (Figure 5).

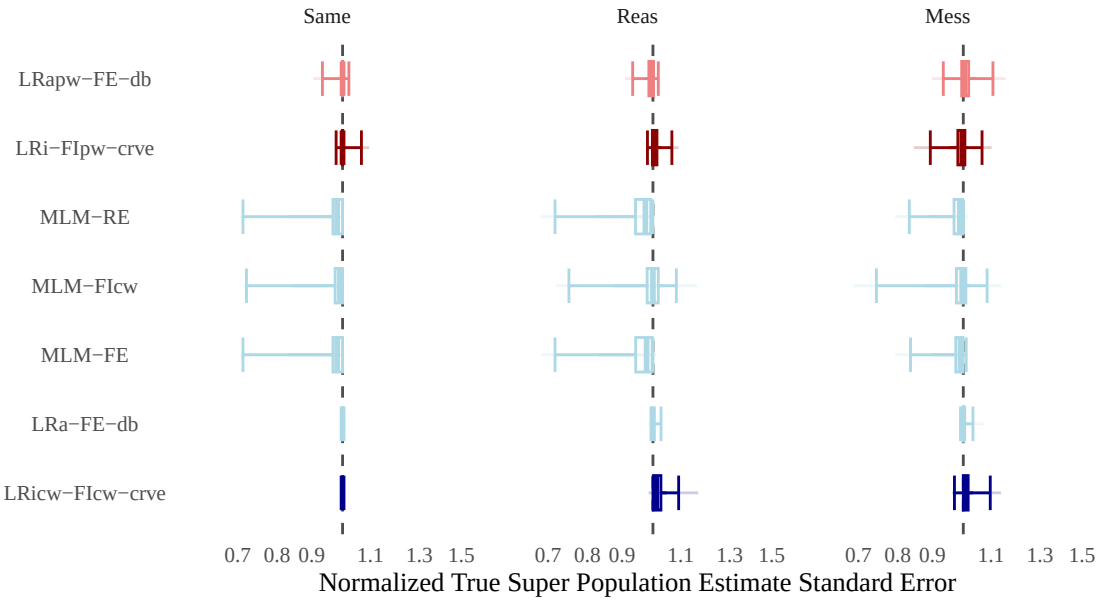


Figure 4: Boxplot of normalized estimate true standard errors across the different scenarios by method. The normalized true standard error is the true standard error for a given method divided by the median true standard error across methods within a given simulation scenario. Therefore each data point is adjusted for the underlying base true standard error from a given set of DGPs and represents the true standard error driven by the estimation method. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

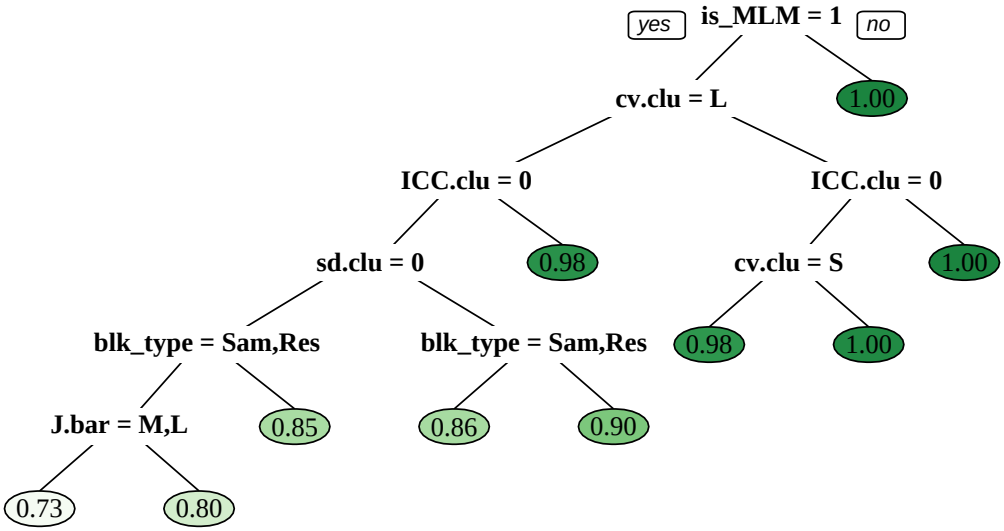


Figure 5: Regression tree predicting the normalized true estimate standard error under different data generating parameters

7 Which estimators have calibrated estimates of the treatment effect standard error?

Overall, across scenarios, the standard error estimates for fully interacted estimators, except the fully interacted MLM, are too small relative to the true super population standard errors (Figure 6). However, across almost all simulations, the magnitude of the underestimate estimate is very small. The only exception is the fully interacted CRVE estimators in some simulations produced severely underestimated standard errors. These were all simulation conditions that likely produced blocks without at least one treated and one control

cluster, or where either all the treatment or control clusters will be very small (Figure 7). This is part of a broader caveat, which is that almost all the fully interacted estimators are undefined in cases where there are only two clusters in any block. While, the fully interacted CRVE estimators do have defined standard error estimates in these cases, the low degrees of freedom simulations confirmed that these estimates are extremely poor and these estimators should not be used under these circumstances.

On the other hand, the estimated standard errors for the other estimators are on average systematically too high across the simulations. For the MLM, the amount of overestimation is small except when there is a confluence of an unbalanced sample, zero cluster-level ICC, and cross-cluster treatment effect variance (Figure 8).

For the fixed effect estimators, the standard errors are well estimated in the same and reasonable blocks conditions. However, we find the standard errors were poorly estimated in the messy blocks, especially when there was no cluster-level ICC were the standard errors are predicted to be overestimated by at least 15% (Figure 9). Although again it is important to remember that the messy blocks condition was designed to test when the estimators break and are not conditions we expect to see in practice. Within these scenarios (i.e. messy blocks and no cluster level ICC), the person weighted estimates were even more overestimated, with this overestimation problem mostly getting worse the less variation there was across clusters.

Consistent with the standard errors being generally well estimated across simulation conditions and estimators, the coverage rate is close to 95% across simulation conditions and estimators (Figure 9). For the MLM models, coverage could fall below 90% under some conditions (e.g., no variation in cluster size, no cluster-level ICC, and at least three other factors) (Figure 10). Under these conditions the weighting for the MLM models shifts from being cluster weighted to person weighted, which causes these estimators to be biased against the cluster weighted estimand; which in turn causes the coverage rate to drop. For the CRVE estimators, the coverage drops predictable under the same conditions that the standard error estimates get worse. Otherwise, there were generally no issues with confidence interval coverage.

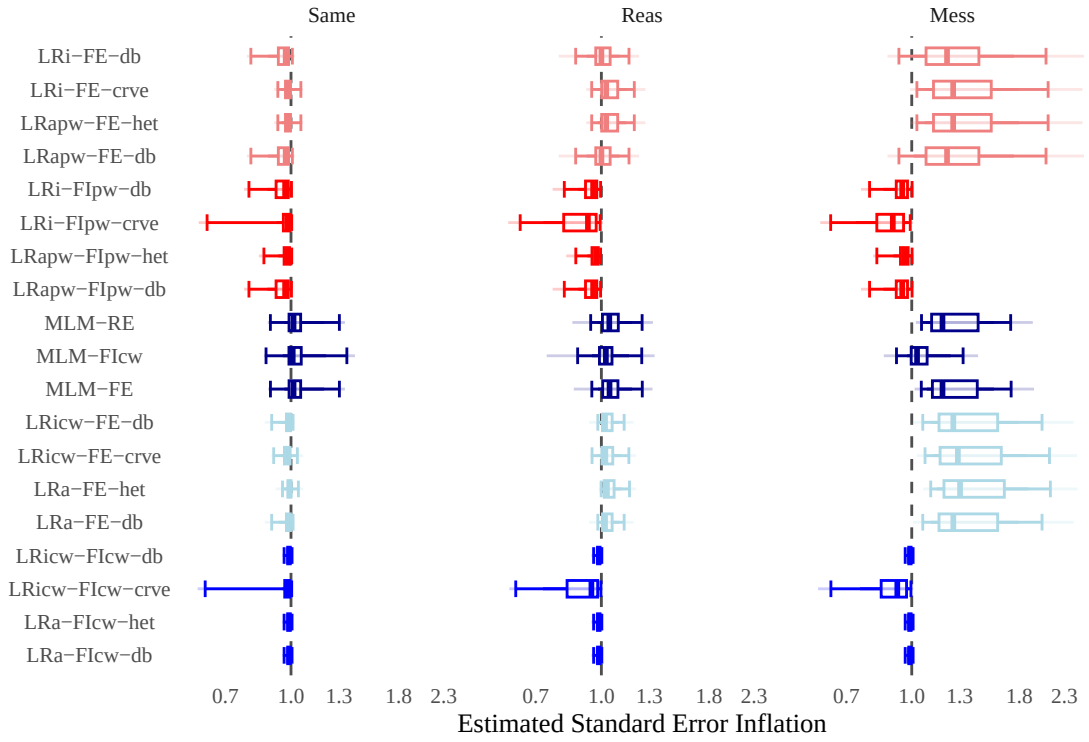


Figure 6: Boxplot of estimated standard error inflation across different simulation scenarios by method. Standard error inflation is measured as the ratio of the estimated standard error over the true super population standard error. Values above 1 indicate overestimation, values below 1 indicate underestimation. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

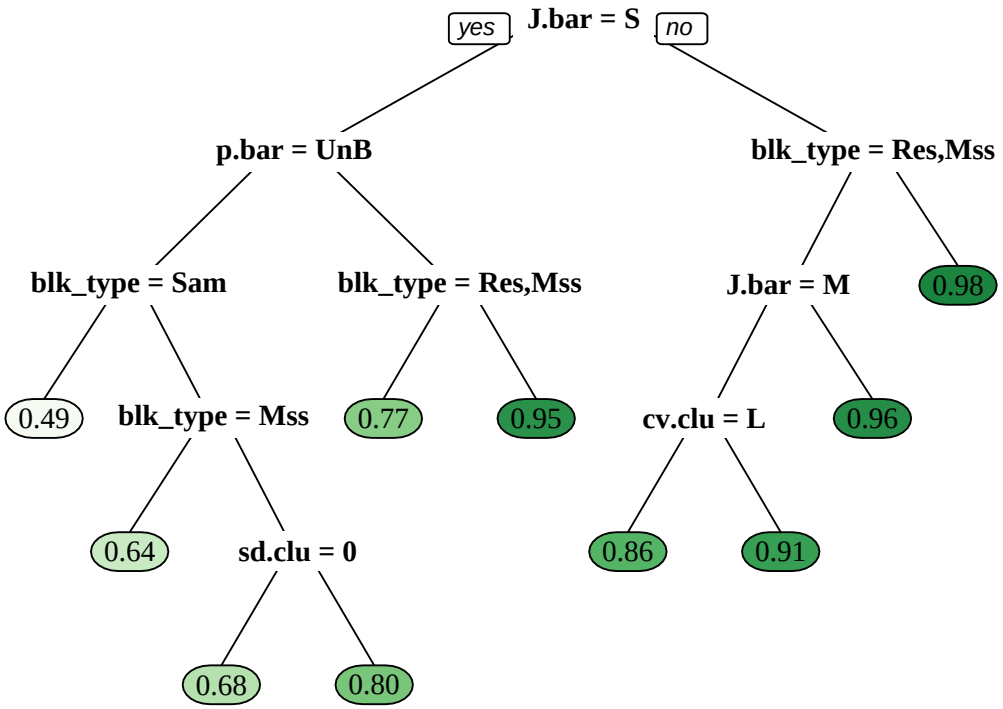


Figure 7: Regression tree which predicts the standard error inflation under different data generating parameters for the fully interacted CRVE estimators

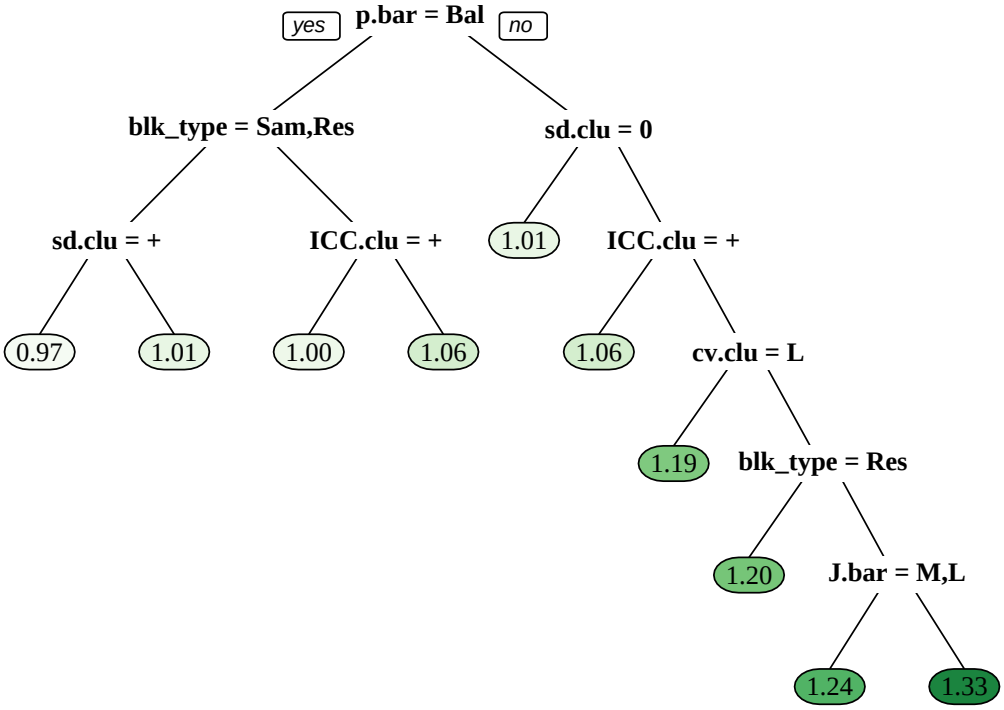


Figure 8: Regression tree which predicts the standard error inflation under different data generating parameters for the multi−level model methods

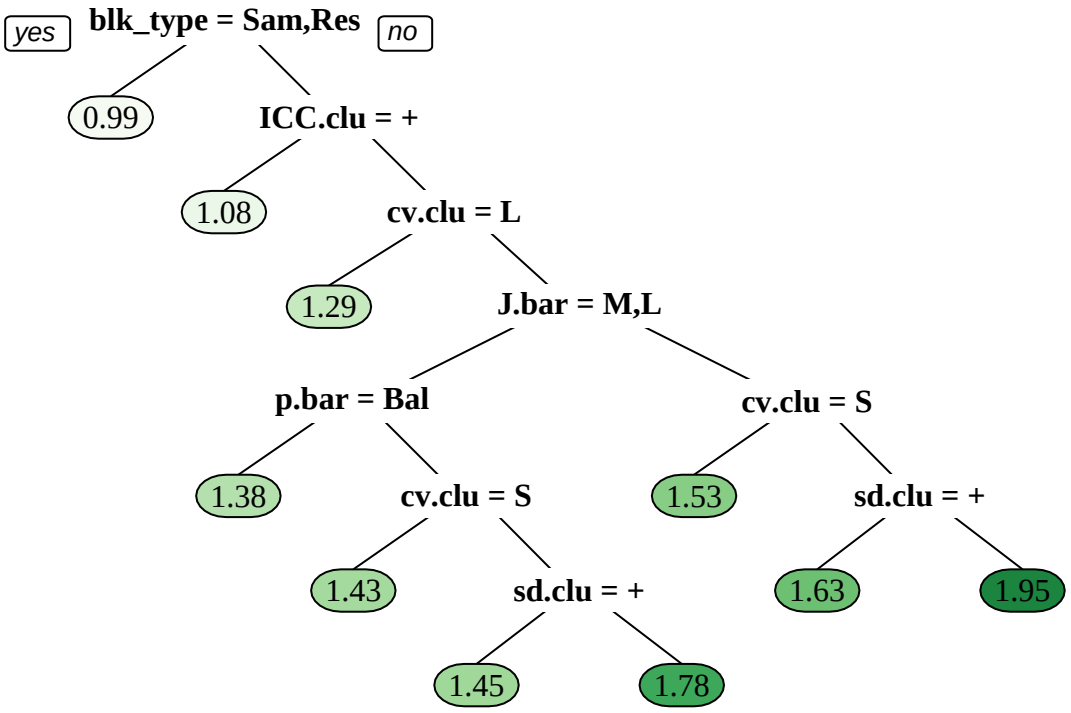


Figure 9: Regression tree which predicts the standard error inflation under different data generating parameters for the fixed effect methods

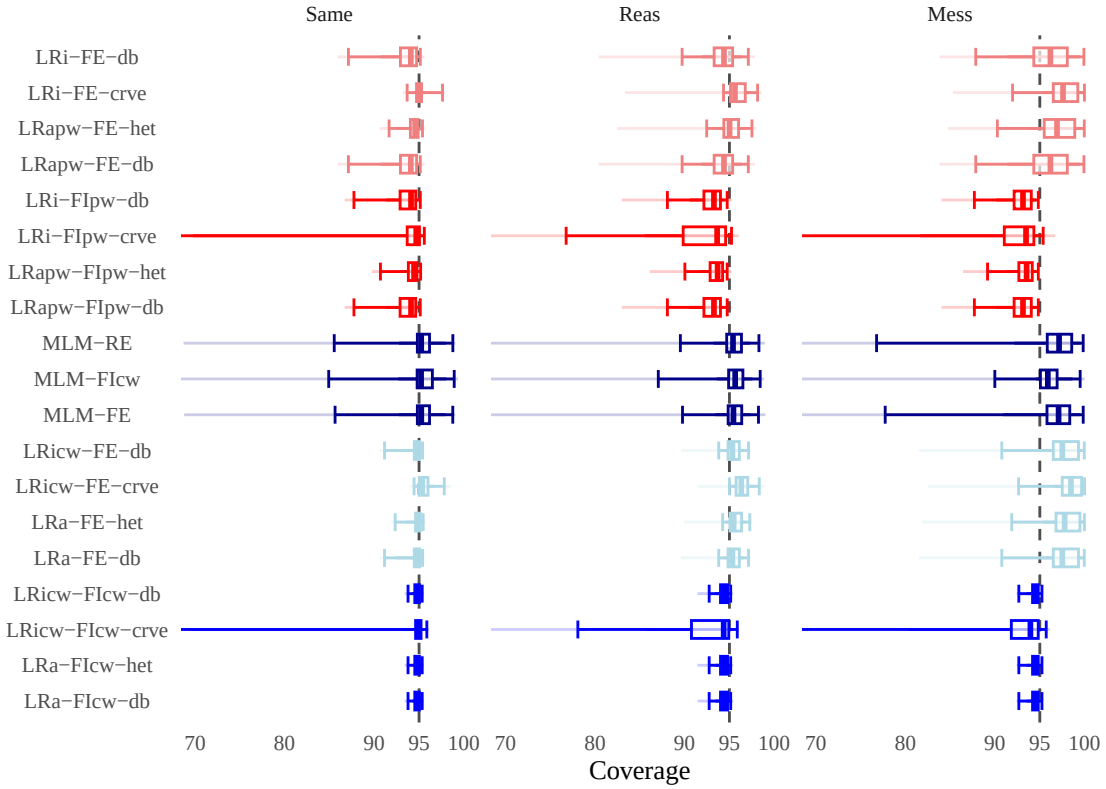


Figure 10: Boxplot of coverage rates across different scenarios by method. The vertical line indicates the target 95% coverage rate. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

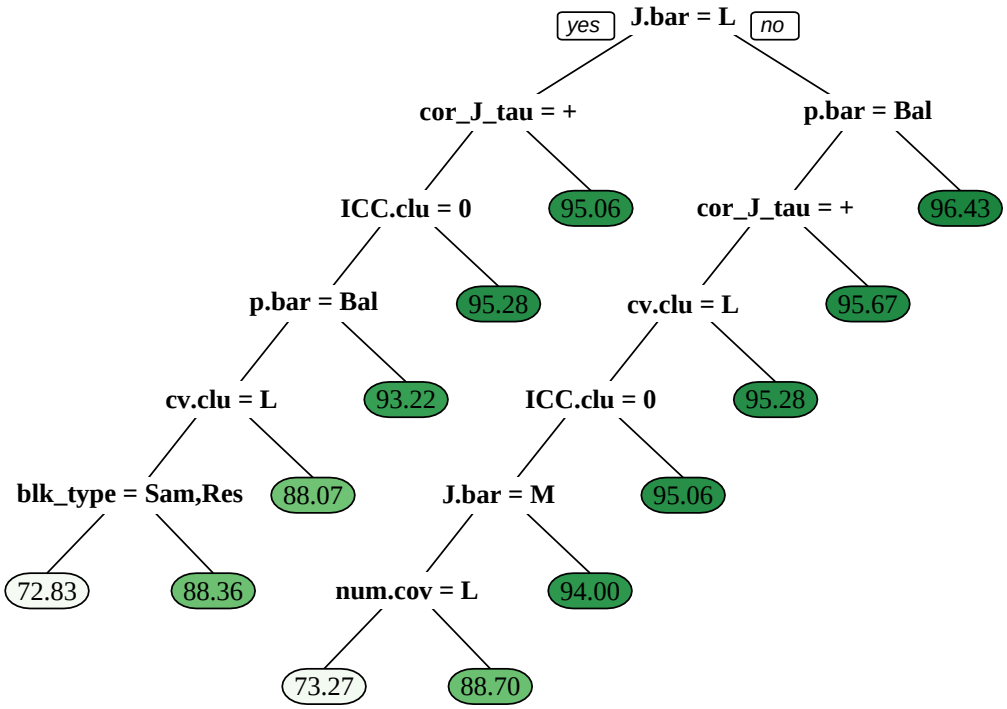


Figure 10: Regression tree which predicts the coverage rates under different data generating parameters for multi−level model methods

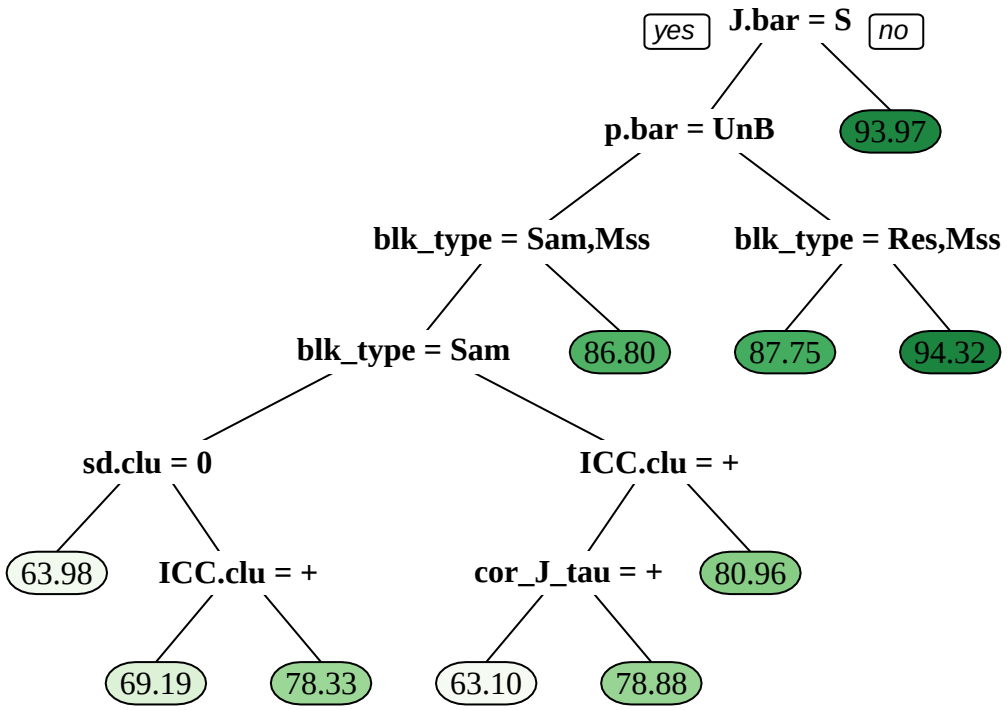


Figure 11: Regression tree which predicts the coverage rates under different data generating parameters for CRVE estimators

8 Do any of the estimated standard errors target finite versus super population uncertainty?

The simulations demonstrate that the standard errors are targeting the super-population standard errors (with blocks being fixed, clusters being sampled for block specific superpop). The true super-population standard errors are only noticeably different from the true finite population standard errors when the cluster-level ICC is zero and there is cross-cluster treatment effect variance (Figure 11). Looking at the subset of simulations where that is true shows that the standard error inflation is consistently closer to one for the true super-population standard errors than for the true finite population standard errors (Figure 12).

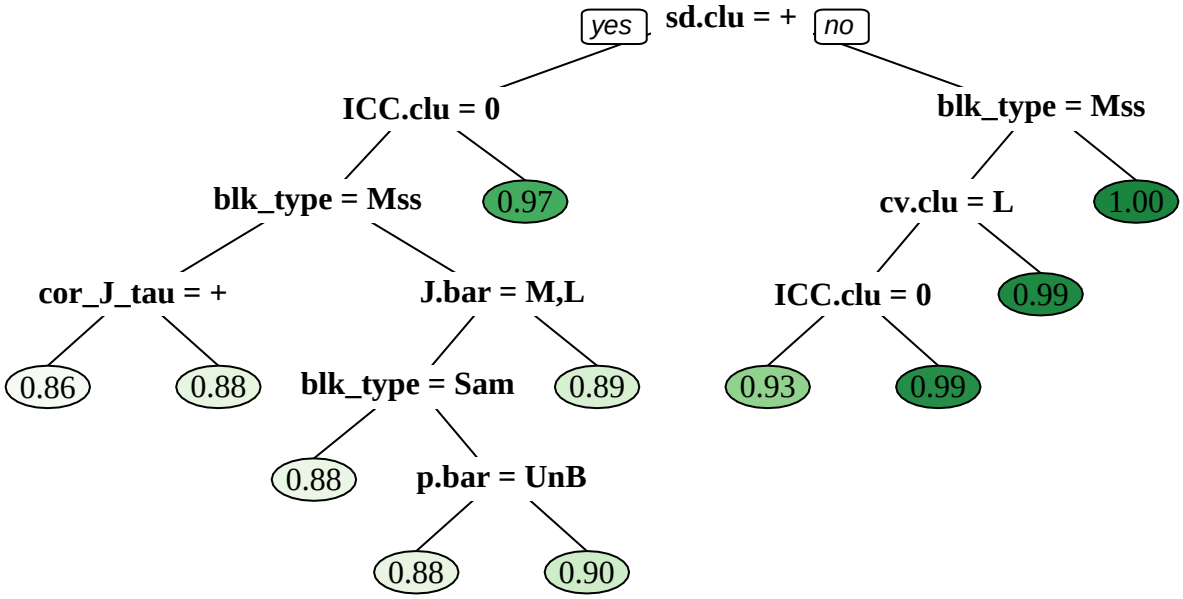


Figure 12: Regression tree which predicts the ratio of true finite population standard error over true super population standard error under different data generating parameters. These two quantities are the same when this ratio is one.

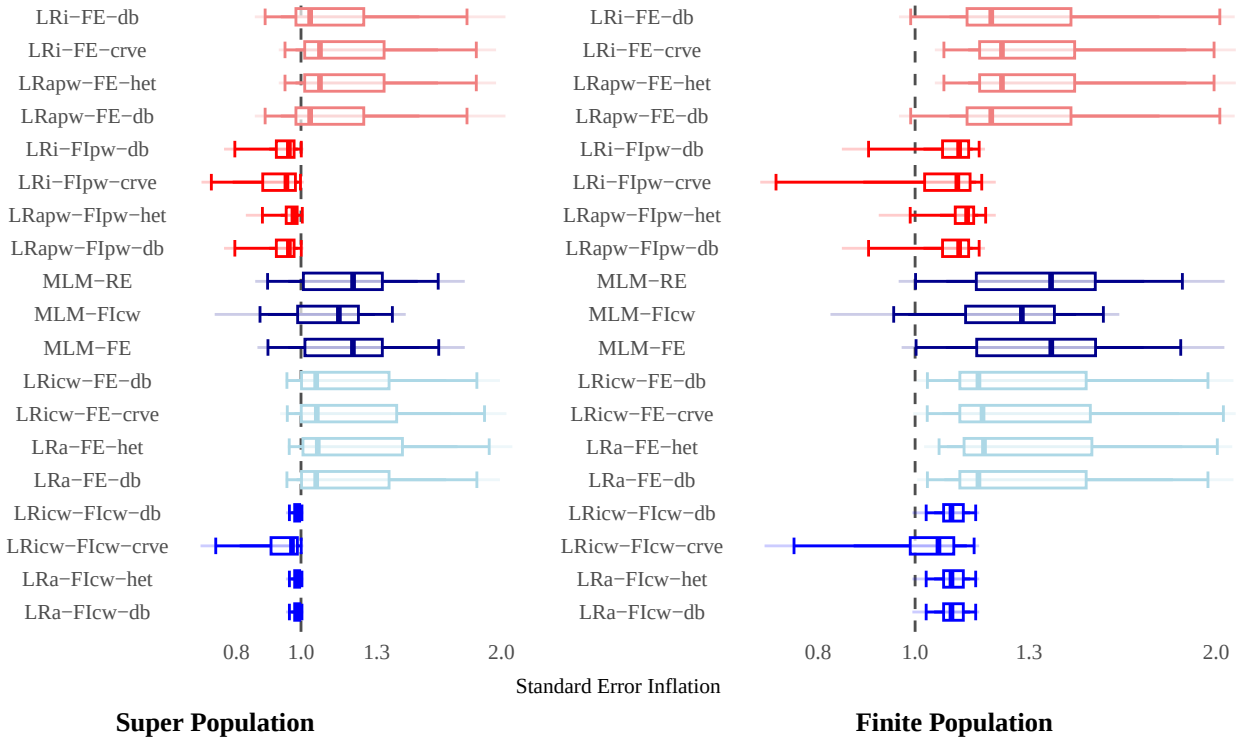


Figure 13: Boxplot of estimated standard error inflation for across simulation scenarios with no cluster−level ICC and non−zero treatment effect heterogeneity by method, comparing againt the true super population and true finite population standard errors. Standard error inflation is measured as the ratio of the estimated standard error over the true standard error. The faint horizontal lines extend to the full range, the bars span the 2.5th to 97.5th percentiles, and the boxes span the 10th to 90th percentiles.

9 Why does the range in the estimates produced by the different estimators get larger as the degrees of freedom get smaller? Is the cause estimator bias or estimator instability?

In the empirical studies the range of estimates produced by different estimators was larger for the low degrees of freedoms studies. Simulations show that this is because the estimates get are more unstable when the degrees of freedom is smaller and not because of anything to do with estimate bias. In the simulations as the degrees of freedom get smaller, the estimate instability gets larger (Figure 13). On the other hand, we see no relationship between mean bias and degrees of freedom in a given scenario (Figure 14).

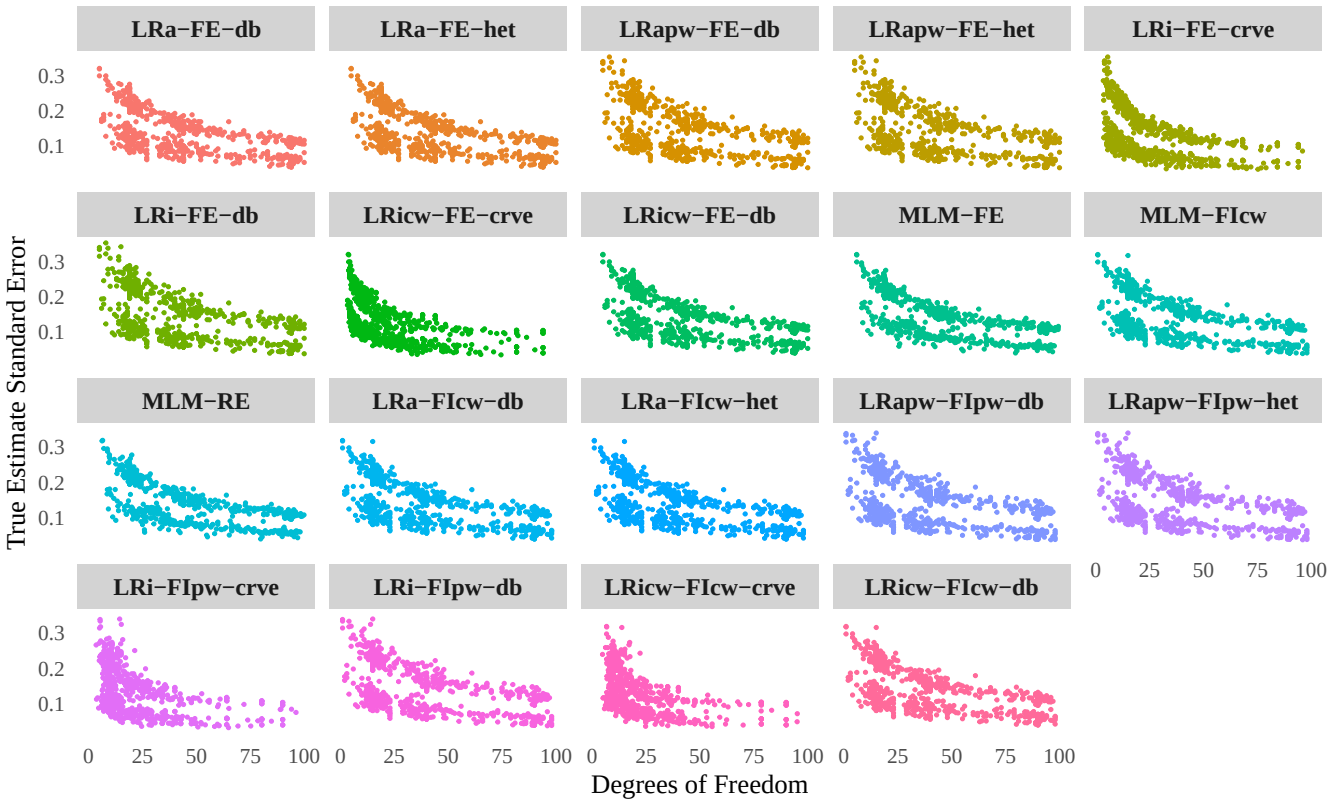


Figure 14: Scatter plot of true estimate standard error versus degrees of freedom by method. Each dot is a given DGP scenario and each panel shows the relationship for a different estimation method.

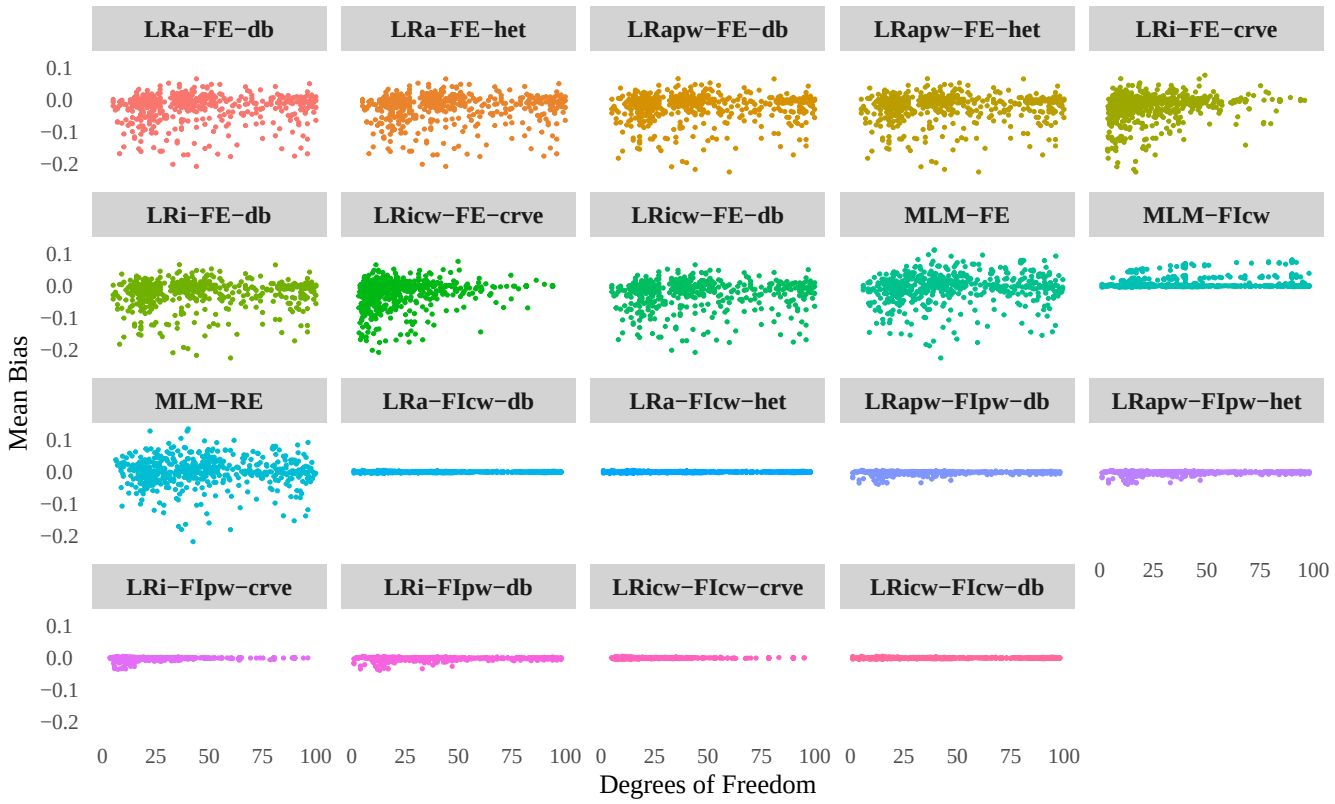


Figure 15: Scatter plot of mean bias versus degrees of freedom by method. Each dot is a given DGP scenario and each panel shows the relationship for a different estimation method.

10 Summary of Results

Table 6: Summary of Simulation Results - Cluster Weighted

Method	Treatment Effect	Standard Errors
Cluster Weighted		
Multi-Level Model	Bias can exceed .04 in simulations when there was a combination of correlation between the number of clusters and the treatment effect, zero cluster-level ICC, and a lot of variation in cluster size In the non-fully interacted MLM models, bias also can exceed .04 when there are messy blocks. Bias variation trade off because the multilevel estimators become more precise than the other estimators when there is a combination of large variation in cluster size and zero cluster ICCs	The standard errors were substantially overestimated in simulations with a confluence of an unbalanced sample, zero cluster-level ICC, and cross-cluster treatment effect variance
Fully Interacted	Minimal bias under all conditions	Except CRVE, very small under estimate of the standard errors under all conditions CRVE standard errors are poorly estimated with a small number of clusters and an unbalanced sample Can not be used if any of the blocks only have two clusters
Fixed Effect	Bias can exceed .05 in simulations when there is a combination of messy block and unbalanced proportion of treated and control clusters	Standard errors were substantially overestimated for messy blocks and no cluster-level ICC

Table 7: Summary of Simulation Results - Person Weighted

Method	**Treatment Effect**	**Standard Errors**
Person Weighted		
Fully Interacted	Minimal bias under all conditions	Except CRVE, very small under estimate of the standard errors under all conditions CRVE standard errors are poorly estimated with a small number of clusters and an unbalanced sample Can not be used if any of the blocks only have two clusters
Fixed Effect	Bias can exceed .05 in simulations when there is a combination of messy block and unbalanced proportion of treated and control clusters	Standard errors were substantially overestimated for messy blocks and no cluster-level ICC